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$$\text{Re}(u) = \cos(k_x x + k_y y - \omega t)$$

$$\text{Im}(u) = \sin(k_x x + k_y y - \omega t)$$

These satisfy wave equation, so

$$\frac{\partial \text{Re}(u)}{\partial t} = (-\omega)(-\sin(k_x x + k_y y - \omega t)) = \omega \sin(k_x x + k_y y - \omega t)$$

$$\frac{\partial^2 \text{Re}(u)}{\partial t^2} = -\omega^2 \text{Re}(u)$$

$$c^2 \nabla^2 \text{Re}(u) = c^2 (-k_x^2 \text{Re}(u) - k_y^2 \text{Re}(u))$$

So because wave eq. holds: $\omega^2 \text{Re}(u) = c^2 (k_x^2 + k_y^2) \text{Re}(u)$

$$\Rightarrow \omega^2 = c^2 (k_x^2 + k_y^2)$$

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$$u(t, x, y) = \exp[i(k_x x + k_y y - \omega t)]$$

$$\text{L.H.S: } \frac{\partial u}{\partial t} = -i\omega u$$

$$\frac{\partial^2 u}{\partial t^2} = \cancel{(-i\omega)^2} - (-i\omega)^2 u = -\omega^2 u$$

$$\text{R.H.S: } \nabla^2 u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}$$

$$\frac{\partial u}{\partial x} = ik_x u \quad \frac{\partial^2 u}{\partial x^2} = (ik_x)^2 u = -k_x^2 u$$

$$\text{By symmetry: } \frac{\partial^2 u}{\partial y^2} = -k_y^2 u$$

$$\text{So R.H.S} = c^2 \nabla^2 u = -c^2 u (k_x^2 + k_y^2),$$

And since we found that $c^2(k_x^2 + k_y^2) = \omega$,

~~we~~ we get L.H.S = R.H.S

$$\exp[i\tilde{\omega}\Delta t n] -$$

$$\frac{u_{ij}^n}{h^2}$$

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$$u_{(i+s)j}^n = \exp[i(kh(i+j) + khs - \tilde{\omega}n\Delta t)] \\ = u_{ij}^n \cdot \exp(ikhs)$$

$$\text{Therefore, } u_{i+1j}^n - 2u_{ij}^n + u_{i-1j}^n \\ = u_{ij}^n (\exp(ikh) + \exp(-ikh) - 2)$$

Second term becomes the same, so

$$\text{RHS} = \frac{c^2}{h^2} 2u_{ij}^n (e^{ikh} + e^{-ikh} - 2)$$

$$c = \frac{1}{\sqrt{2}} = \frac{c \Delta t}{h}$$

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$$u_{ij}^n = \exp[i(kh(i+j) - \tilde{\omega} n \Delta t)] = \exp[ikh(i+j)] \cdot \exp[-i\tilde{\omega} \Delta t n]$$

$$u_{ij}^{n+2} - 2u_{ij}^n = u_{ij}^{n+k} = \exp(ikh(i+j)) \cdot \exp(-i\tilde{\omega} \Delta t n) \cdot \exp(-i\tilde{\omega} \Delta t k)$$

So LHS becomes

$$\begin{aligned} & \exp(ikh(i+j)) \cdot \exp(-i\tilde{\omega} \Delta t n) \cdot [\exp(-i\tilde{\omega} \Delta t) - 2 + \exp(i\tilde{\omega} \Delta t)] \cdot \frac{1}{\Delta t^2} \\ &= \frac{u_{ij}^n}{\Delta t^2} \cdot (e^{i\tilde{\omega} \Delta t} + e^{-i\tilde{\omega} \Delta t} - 2) \end{aligned}$$

$$\begin{aligned} \bullet \text{ RHS} = \text{LHS} & \Leftrightarrow \frac{2c^2}{h^2} (e^{ikh} + e^{-ikh} - 2) = \frac{1}{\Delta t^2} (e^{i\tilde{\omega} \Delta t} + e^{-i\tilde{\omega} \Delta t} - 2) \\ & \Leftrightarrow 2c^2 \cdot (e^{ikh} + e^{-ikh} - 2) = e^{i\tilde{\omega} \Delta t} + e^{-i\tilde{\omega} \Delta t} \end{aligned}$$

Use $c = \frac{1}{\sqrt{2}}$ and $e^{ix} + e^{-ix} = 2\cos x$, we get

$\cos(\tilde{\omega} \Delta t) = \cos(kh)$. Looking at the principle branch, we get

$$\tilde{\omega} \Delta t = kh \Rightarrow \tilde{\omega} = \frac{kh}{\Delta t}$$

Since $c = \frac{c \Delta t}{h} = \frac{1}{\sqrt{2}}$, we get $\Delta t = \frac{h}{c\sqrt{2}}$ and therefore

$\tilde{\omega} = c\sqrt{2}k$. From 1.2.3) we have $\omega = c\sqrt{2}k$,

so finally we arrive at $\tilde{\omega} = \omega$ $\square$